

Course Logistics

CMSC477 – Robotics: Perception & Planning
Spring 2026

TAs: Anukriti Singh, Khuzema Habib

Lab Manager: Ivan Pensiky

Agenda

- Course Website/ELMS
- Lab Logistics
- Robots
- Getting Started: Project Zero
- Homework 1



Discussions on ELMS

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Spring 2026

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Discussions

Grades

People

Syllabus

BigBlueButton

Collaborations

Chat

Clickers

Course Reserves

Adobe Creative Cloud

Top Hat

Lucid (Whiteboard)

All



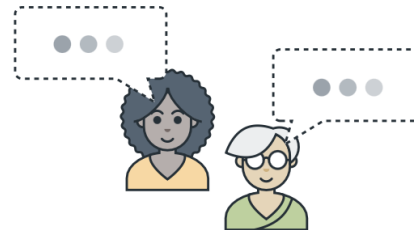
🔍 Search by title or author...

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[Click here to add a discussion](#)

Labs

- Lab, 3rd Floor of IDEA Factory
 - Managed by Ivan Pensiky
- Official Lab Hours (attendance mandatory)
 - Section 0101: Monday 2:00 - 4:50 pm
 - Section 0202: Tuesday 2:30 - 5:20 pm

Labs

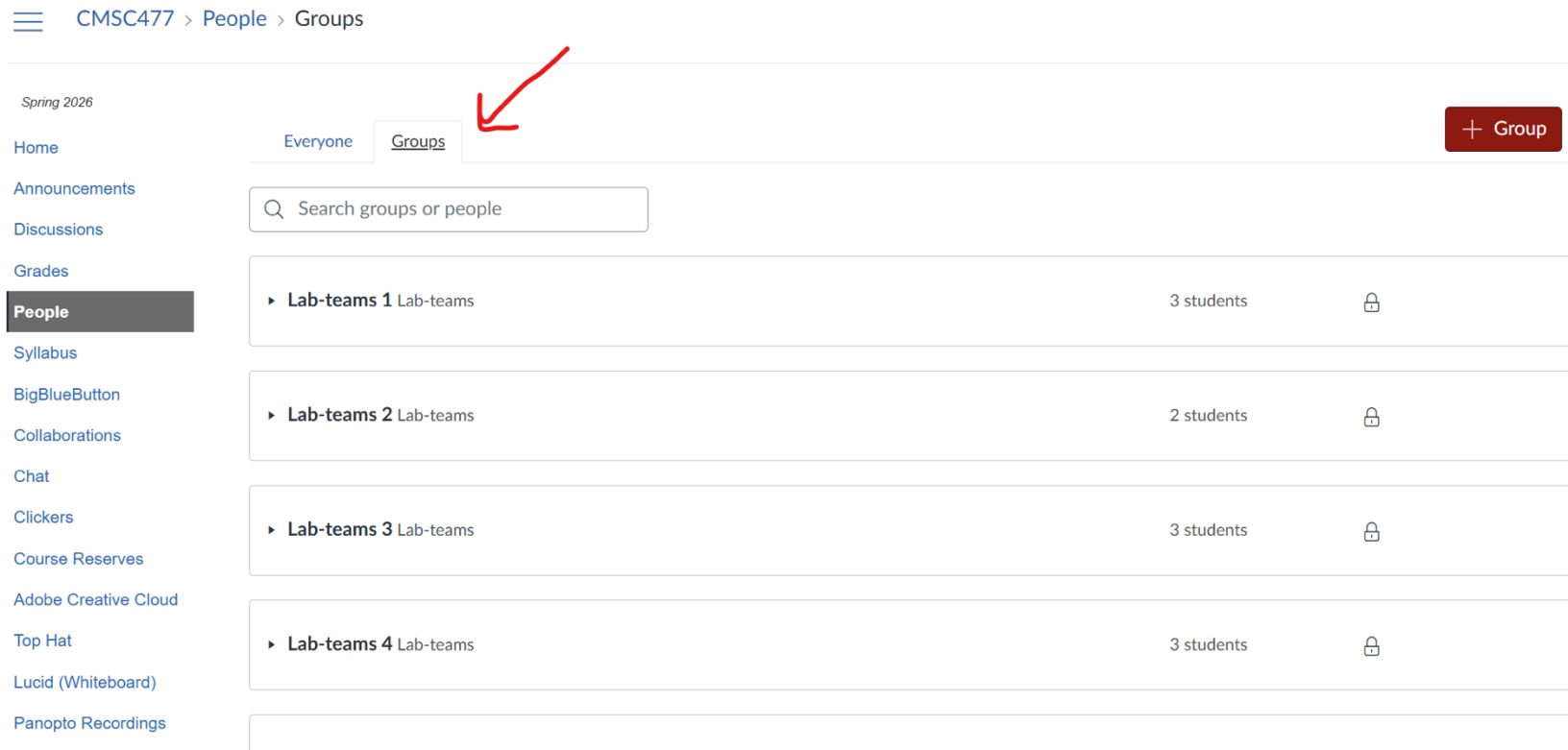
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- ***Who is YOUR TA?***

Labs

- Lab, 3rd Floor of IDEA Factory
 - Managed by Ivan Pensiky
- Official Lab Hours (attendance mandatory)
 - Section 0101: Monday 2:00 - 4:50 pm — **Khuzema Habib**
 - Section 0202: Tuesday 2:30 - 5:20 pm — **Anukriti Singh**
- Direct any questions about the assignments, labs, and robots to your TA!

Team Assignments

- List is be posted on ELMS



The screenshot shows the ELMS interface for CMSC477. The breadcrumb trail is CMSC477 > People > Groups. The left sidebar contains various navigation links, with 'People' highlighted. The main content area shows the 'Groups' tab selected, indicated by a red arrow. Below the tab is a search bar labeled 'Search groups or people'. A table lists four lab teams, each with a name, student count, and a lock icon.

Group Name	Students	Lock Icon
Lab-teams 1 Lab-teams	3 students	🔒
Lab-teams 2 Lab-teams	2 students	🔒
Lab-teams 3 Lab-teams	3 students	🔒
Lab-teams 4 Lab-teams	3 students	🔒

Team Assignments

- List is be posted on ELMS
- You are free to switch teams
 - As long as number of teams does not change!
 - Let Anukriti know about any changes by Friday, Feb 5—*otherwise, you will remain with the same individuals you were assigned!*
- One (1) robot per team
 - Your team's robot id numbers are on the sheet

End of Semester Competition Date

- May 6th is problematic.
- Other options for competition dates:
 - May 4
 - Final Exam day (May 14)

Let's Meet Your TAs

Khuzema Habib --- Mondays (Section 0101)

Anukriti Singh --- Wednesdays (Section 0102)

Robots

- Each team has one (1) DJI RoboMaster
 - Each robot has 2 batteries
- Camera
- 2 axis Gripper
- Mecanum wheels
 - Can rotate and translate independently
- Communicates via WiFi



Robots

- Python code runs on your laptop
 - Receives camera images
 - Sends motor commands
- Communication occurs over WiFi



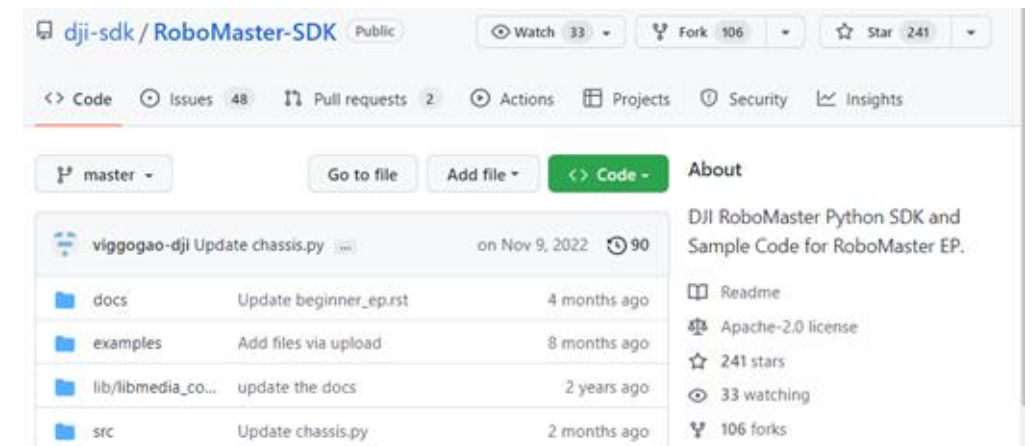
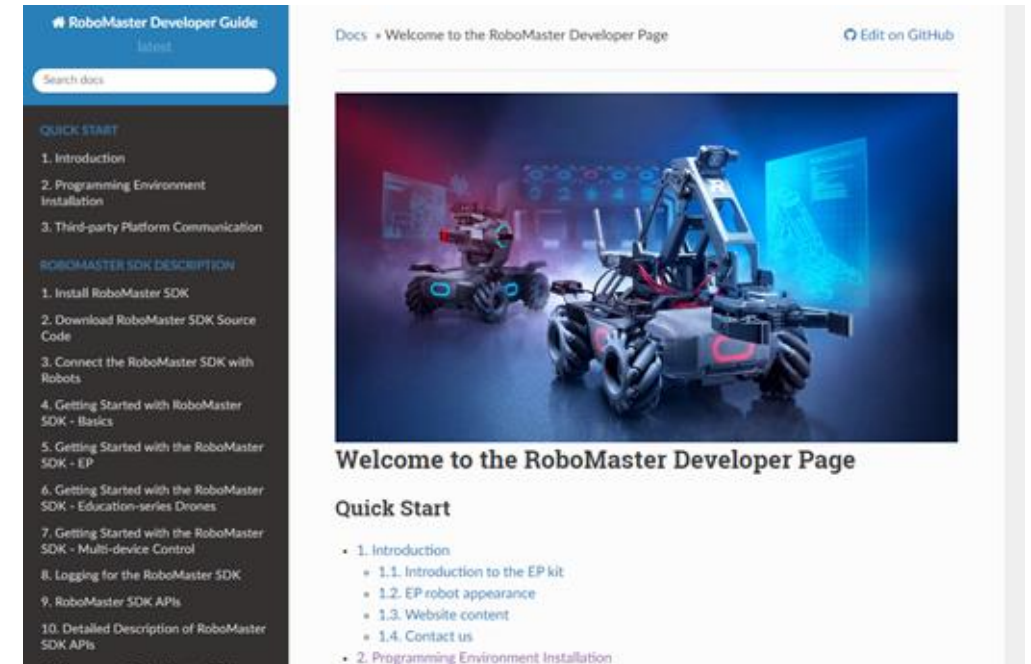
Logistics

- You ***must*** work with your robots in the lab ***only***.
- Be very **careful** with them!
 - They are expensive and hard to replace.
- **Do not** use them on a table.
- **Do not** crash your robot into things!



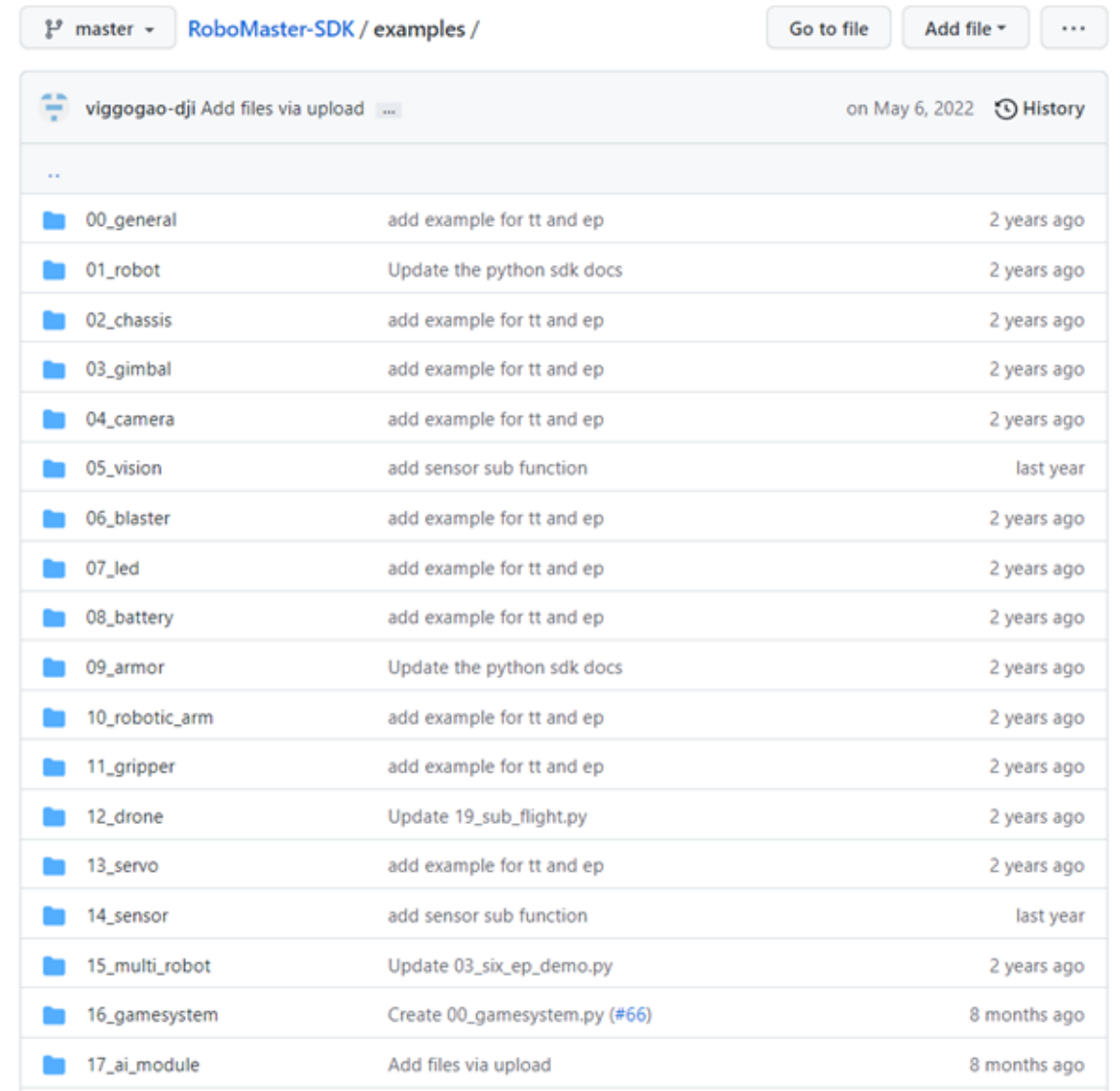
RoboMaster SDK

- Python Library provided by DJI
- Documentation
 - <https://robomaster-dev.readthedocs.io/en/latest/>
- Github Repository
 - <https://github.com/dji-sdk/RoboMaster-SDK>



RoboMaster SDK

- Numerous Python examples



The screenshot shows the GitHub repository for RoboMaster-SDK, specifically the 'examples' directory. The repository is on the 'master' branch. The user 'viggogao-dji' is shown with the option to 'Add files via upload'. The table lists 17 example folders, each with a description of the changes and the time since the last commit.

RoboMaster-SDK / examples /		Go to file	Add file ▾	...
viggogao-dji Add files via upload ...		on May 6, 2022 History		
..				
00_general	add example for tt and ep	2 years ago		
01_robot	Update the python sdk docs	2 years ago		
02_chassis	add example for tt and ep	2 years ago		
03_gimbal	add example for tt and ep	2 years ago		
04_camera	add example for tt and ep	2 years ago		
05_vision	add sensor sub function	last year		
06_blaster	add example for tt and ep	2 years ago		
07_led	add example for tt and ep	2 years ago		
08_battery	add example for tt and ep	2 years ago		
09_armor	Update the python sdk docs	2 years ago		
10_robotic_arm	add example for tt and ep	2 years ago		
11_gripper	add example for tt and ep	2 years ago		
12_drone	Update 19_sub_flight.py	2 years ago		
13_servo	add example for tt and ep	2 years ago		
14_sensor	add sensor sub function	last year		
15_multi_robot	Update 03_six_ep_demo.py	2 years ago		
16_gamesystem	Create 00_gamesystem.py (#66)	8 months ago		
17_ai_module	Add files via upload	8 months ago		

RoboMaster SDK

- Simple API

```
17 import time
18 from robomaster import robot
19
20
21 if __name__ == '__main__':
22     ep_robot = robot.Robot()
23     ep_robot.initialize(conn_type="sta")
24
25     ep_chassis = ep_robot.chassis
26
27     x_val = 0.5
28     y_val = 0.3
29     z_val = 30
30
31     # 前进 3秒
32     ep_chassis.drive_speed(x=x_val, y=0, z=0, timeout=5)
33     time.sleep(3)
34
35     # 后退 3秒
36     ep_chassis.drive_speed(x=-x_val, y=0, z=0, timeout=5)
37     time.sleep(3)
38
39     # 左移 3秒
40     ep_chassis.drive_speed(x=0, y=-y_val, z=0, timeout=5)
41     time.sleep(3)
42
43     # 右移 3秒
44     ep_chassis.drive_speed(x=0, y=y_val, z=0, timeout=5)
45     time.sleep(3)
46
47     # 左转 3秒
48     ep_chassis.drive_speed(x=0, y=0, z=-z_val, timeout=5)
49     time.sleep(3)
50
51     # 右转 3秒
52     ep_chassis.drive_speed(x=0, y=0, z=z_val, timeout=5)
53     time.sleep(3)
54
55     # 停止麦轮运动
56     ep_chassis.drive_speed(x=0, y=0, z=0, timeout=5)
```


Project 0

- Description posted today on website
- Track an AprilTag that is moving
 - Remain a fixed distance away
- This challenge is just to get you started
- Video due: **February 20th**
 - *Two* full lab sessions till due date
 - Later projects will have written report



Homework 1

- One and only homework in this class
- Will be released by February 10th

CMSC477: Robotics Perception and Planning Homework 1: Graph-Based Planning

Dr. Troi Williams, Anukriti Singh, Khuzema Habib

February 10, 2026

1 Introduction

In this assignment, you will learn to find the shortest valid path for a robot to move from a source to a destination, given a series of obstacles. You will be solving this problem using Graph-Based Search Algorithms:

- Breadth-First Search (BFS)
- Depth-First Search (DFS)
- Dijkstra

Questions?